

ABPNoiseSweep Run Manual

Data-only ABP/MUCA cluster workflow: setup, configuration, output locations, and saved data.

Scope. This manual describes the cleaned project layout in **ABPNoiseSweep_project_clean_v2.zip**. It assumes plotting is done by separate code. The simulation writes JLD2 result files and lightweight CSV exports.

1. Project structure

After unzipping, the project should look like this:

```
ABPNoiseSweep/
|-- Project.toml           # Julia environment/dependencies
|-- README.md
|-- src/                   # reusable simulation code
|   |-- config.jl          # default parameters and validation
|   |-- cli.jl             # command-line/environment interface
|   |-- driver.jl          # high-level sweep loop
|   |-- model.jl           # ABP dynamics and observables
|   |-- moves.jl           # MCMC proposal moves
|   |-- muca.jl            # recursive MUCA learning
|   |-- production.jl      # production sampling
|   |-- io.jl              # JLD2 and CSV output
|-- scripts/run_noise_sweep.jl # Julia entry point
|-- run/run_abp_noise_sweep_data_only.sh # cluster/Slurm launcher
|-- test/runtests.jl
`-- docs/cluster_usage.md
```

The intended execution path is:

```
.sh launcher
-> scripts/run_noise_sweep.jl
-> src/cli.jl builds the config
-> src/driver.jl loops over D values
-> src/muca.jl + src/production.jl run the simulation
-> src/io.jl writes JLD2 and CSV data
```

2. First setup from zero

Run these commands from the directory where you placed the ZIP file.

```
unzip ABPNoiseSweep_project_clean_v2.zip
cd ABPNoiseSweep
julia --project=. -e 'import Pkg; Pkg.instantiate()'
julia --project=. -e 'import Pkg; Pkg.test()'
```

If the cluster requires modules, load Julia first, for example `module load julia`. The exact module name depends on the cluster.

3. Check the configuration before running

Use a dry run whenever you change noise levels or iteration settings. It prints the final config and the effective MUCA iteration count for each D, then stops before simulation.

```
julia --project=. scripts/run_noise_sweep.jl --dry-run
```

```
# Or through the cluster launcher:
./run/run_abp_noise_sweep_data_only.sh --dry-run
```

Use a smoke test to check that the code launches and writes small output files:

```
./run/run_abp_noise_sweep_data_only.sh --smoke
```

4. Full cluster submission

Run this from inside ABPNoiseSweep. This is one terminal command. Do not include an extra period at the end. The line-broken form below is safer to read:

```
sbatch --export=ALL,\  
ABP_D_VALUES=0.1,0.01,0.005,\  
ABP_OUTPUT_DIR=run_name,\  
ABP_MOVE_WEIGHTS=0.0,0.05,0.55,0.40,\  
ABP_N_ITER=90,\  
ABP_N_ITER_STEPS_PER_ITER=120000000 \  
run/run_abp_noise_sweep_data_only.sh
```

Each backslash must be the last character on its line; do not place spaces after it. A one-line command is also valid if your terminal makes it easier.

5. Where the data goes

The output folder is controlled by ABP_OUTPUT_DIR. If you set ABP_OUTPUT_DIR=my_run, outputs go into:

```
my_run/  
|-- abp_endpoint_conditioned_D0p1.jld2  
|-- abp_endpoint_conditioned_D0p01.jld2  
|-- abp_endpoint_conditioned_D0p005.jld2  
`-- data/  
    |-- abp_endpoint_conditioned_D0p1/  
    |   |-- case_summary.csv  
    |   |-- muca_last_histogram_logweights.csv  
    |   |-- muca_iteration_diagnostics.csv  
    |   |-- hist_endpoint_x_T.csv  
    |   |-- endpoint_positive_path_y.csv  
    |   |-- endpoint_positive_path_x_y_long.csv  
    |   |-- conditional*_one_dimensional_histograms.csv  
    |-- abp_endpoint_conditioned_D0p01/  
    |-- abp_endpoint_conditioned_D0p005/  
    |-- roundtrip_scaling_rows.csv  
    `-- roundtrip_scaling_powerLaw_fit.csv
```

Important: use a new ABP_OUTPUT_DIR for every serious run. Otherwise different jobs can write into the same folder.

6. How to change noise levels and number of noise levels

Yes: you can change both the noise values and how many values are swept. In the current model each listed value is used as $D_t = D_r = D$. The number of noise levels is simply the number of comma-separated entries in `ABP_D_VALUES`.

Goal	Command fragment
One noise level	<code>ABP_D_VALUES=0.01</code>
Three noise levels	<code>ABP_D_VALUES=0.1,0.01,0.005</code>
Five noise levels	<code>ABP_D_VALUES=0.1,0.05,0.02,0.01,0.005</code>

Example dry run for five noise values:

```
ABP_D_VALUES=0.1,0.05,0.02,0.01,0.005 \  
ABP_OUTPUT_DIR=test_five_D \  
./run/run_abp_noise_sweep_data_only.sh --dry-run
```

If later you want separate translational and rotational noise sweeps, the code would need a small extension; currently the single `D` value sets both D_t and D_r .

7. How iterations change with noise

The submitted `ABP_N_ITER` is the reference MUCA iteration count. When `ABP_SCALE_N_ITER_WITH_D=true`, the effective count is:

```
n_iter_effective(D) = round(ABP_N_ITER * sqrt(ABP_D_SCALING_REFERENCE / D))
```

With the defaults `ABP_N_ITER=90` and `ABP_D_SCALING_REFERENCE=0.01`:

D	factor $\sqrt{0.01/D}$	effective MUCA iterations
0.1	0.316	28
0.01	1.000	90
0.005	1.414	127

To see the exact values for your chosen list, run:

```
ABP_D_VALUES=0.1,0.01,0.005 \  
ABP_N_ITER=90 \  
ABP_D_SCALING_REFERENCE=0.01 \  
./run/run_abp_noise_sweep_data_only.sh --dry-run
```

To disable `D`-dependent iteration scaling:

```
ABP_SCALE_N_ITER_WITH_D=false ./run/run_abp_noise_sweep_data_only.sh --dry-run
```

8. How to change move weights and reflection

Reflection is not a separate sweep variable. It is only controlled by the first entry of the move-weight vector:

```
ABP_MOVE_WEIGHTS = reflection, theta0, block_noise, local_noise
```

Configuration	Meaning
0.0,0.05,0.55,0.40	Default: no reflection moves
0.05,0.05,0.50,0.40	5 percent reflection, 5 percent θ_0 , 50 percent block, 40 percent local

Proposal sizes are fixed across `D` by default:

```
ABP_BLOCK_DXI=0.05  
ABP_LOCAL_DXI=0.8
```

9. Main configuration variables

Variable	What it changes
ABP_D_VALUES	Comma-separated list of D values; each is used as $D_t=D_r=D$.
ABP_OUTPUT_DIR	Top-level output directory for JLD2 and CSV files.
ABP_N_ITER	Reference MUCA iteration count before D scaling.
ABP_D_SCALING_REFERENCE	Reference D used in the iteration-scaling formula.
ABP_SCALE_N_ITER_WITH_D	true/false: whether to scale n_iter with D.
ABP_N_ITER_STEPS_PER_ITER	Approximate sampling moves in the final MUCA iteration, across chains.
ABP_MOVE_WEIGHTS	Move probabilities: reflection, theta0, block noise, local noise.
ABP_BLOCK_DXI	Block-noise proposal amplitude in standardized noise variables.
ABP_LOCAL_DXI	Local-noise proposal amplitude in standardized noise variables.
ABP_N_PROD_OBS_TOTAL	Total production observations across chains.
ABP_SAVE_CSV	true/false: write lightweight CSV exports.

10. What is saved in JLD2

Each D value produces one JLD2 file. This is the complete result file and should be kept. It contains metadata, MUCA data, production histograms, reweighting diagnostics, roundtrips, conditional-window histograms, and selected saved trajectories.

JLD2 path/prefix	Contents
metadata/*	D, Dt, Dr, v, dt, move weights, n_iter, n_iter_effective, proposal sizes, production settings, timing.
muca/bins_bias, muca/centers_bias	Bias bin edges and centers for endpoint $x(T)$.
muca/last_logw_values	Final MUCA logweights used for reweighting/plotting.
muca/iter_logw_values	Full MUCA logweight history, kept in JLD2 only.
muca/iter_hist_values	MUCA histogram history.
muca/iter_accept	Acceptance diagnostic per MUCA iteration.
muca/iter_roundtrips	Roundtrips counted per MUCA iteration.
muca/iter_steps_per_target_roundtrips	Estimated sampling moves needed for target roundtrips.
histograms/biased/* and histograms/unbiased/*	Biased and reweighted production histograms.
endpoint_positive_condition/*	Whole-path y and x-y data conditioned on $x(T)>0$.
conditional_windows/*	Endpoint-window conditional histograms and diagnostics.
roundtrips/*	Production roundtrip trace and final count.
saved_trajectories/*	Selected example trajectories for inspection.

11. What is saved in CSV

CSV exports are intentionally lighter than JLD2. They are meant for plotting code and quick inspection. Per your preference, CSV contains the final MUCA logweights only, not the full logweight history.

CSV file	Contents
case_summary.csv	Key metadata and diagnostics for one D.
muca_last_histogram_logweights.csv	Bias centers, final MUCA histogram, final logweights, inverse weights relative to x_0 .
muca_iteration_diagnostics.csv	Acceptance, flatness, roundtrips, sampling steps, and roundtrip-rate estimates per iteration.
hist_endpoint_x_T.csv	Biased and unbiased endpoint $x(T)$ histogram/PDF.
endpoint_positive_path_y.csv	Whole-path y histogram/PDF conditioned on $x(T)>0$.
endpoint_positive_path_x_y_long.csv	Long-format x-y path heatmap data conditioned on $x(T)>0$.
conditional_*_one_dimensional_histograms.csv	Conditional y_T , y_{mean} , y_{int} histograms for endpoint windows.
roundtrip_scaling_rows.csv	One row per D with roundtrip scaling diagnostics and effective iteration counts.
roundtrip_scaling_power_law_fit.csv	Simple cross-D power-law fit for roundtrip steps.

12. Recommended workflow for each serious run

```
cd ABPNoiseSweep

# 1) Decide a unique output directory.
export ABP_OUTPUT_DIR=run_YYYYMMDD_short_description

# 2) Set the noise list and important run parameters.
```

```

export ABP_D_VALUES=0.1,0.01,0.005
export ABP_N_ITER=90
export ABP_N_ITER_STEPS_PER_ITER=120000000
export ABP_MOVE_WEIGHTS=0.0,0.05,0.55,0.40

# 3) Verify derived settings before running.
./run/run_abp_noise_sweep_data_only.sh --dry-run

# 4) Optional small test.
./run/run_abp_noise_sweep_data_only.sh --smoke

# 5) Submit the real job.
sbatch --export=ALL run/run_abp_noise_sweep_data_only.sh

```

13. Quick answers to common questions

Question	Answer
Can I change the noise values?	Yes. Set ABP_D_VALUES. The number of entries is the number of D cases.
Does the code save MUCA logweights?	Yes. Final logweights are in CSV and JLD2. Full iteration history is in JLD2.
Does CSV contain all logweight history?	No, by design. CSV stays lighter and contains only final logweights plus iteration diagnostics.
Where are the main result files?	In ABP_OUTPUT_DIR as one .jld2 file per D.
Where are plotting-friendly files?	In ABP_OUTPUT_DIR/data/.
How do I include reflection?	Change the first ABP_MOVE_WEIGHTS entry above zero.
How do I know the actual iterations?	Run --dry-run and read n_iter_effective for each D.

End of manual.